How to Optimise the ERNIE magnet using Compute Canada.

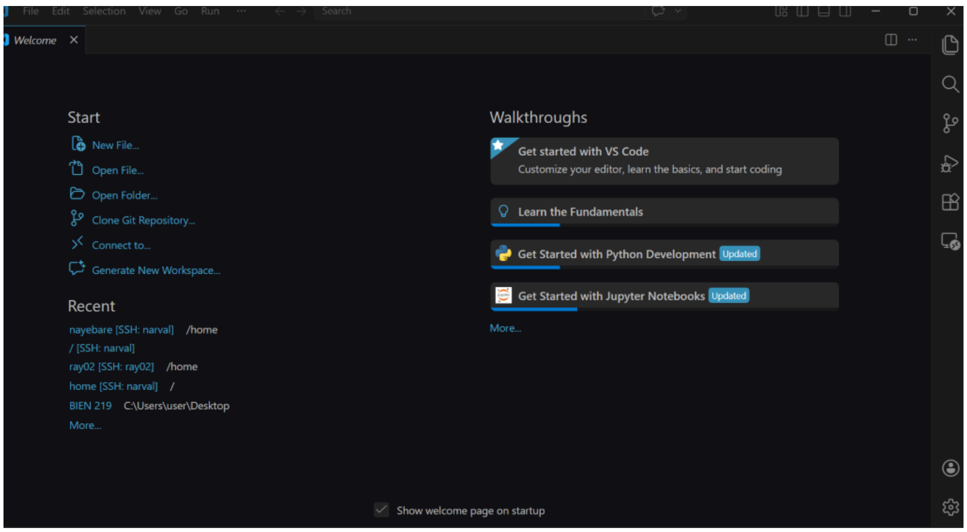
Pre-requisites: Install Visual Studio Code, the appropriate version of Python according to [*https://github.com/tmachtelinckx/halbacharray-GA*](https://github.com/tmachtelinckx/halbacharray-GA)

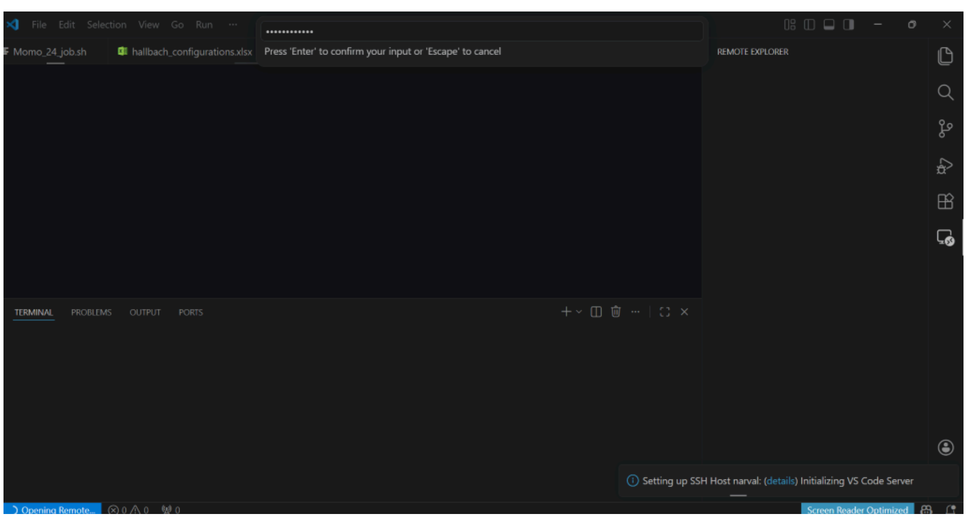
Set up a Compute Canada account.

Step 1: Open VS Code

Step 2: select ssh naval/ home

Step 3: Enter password

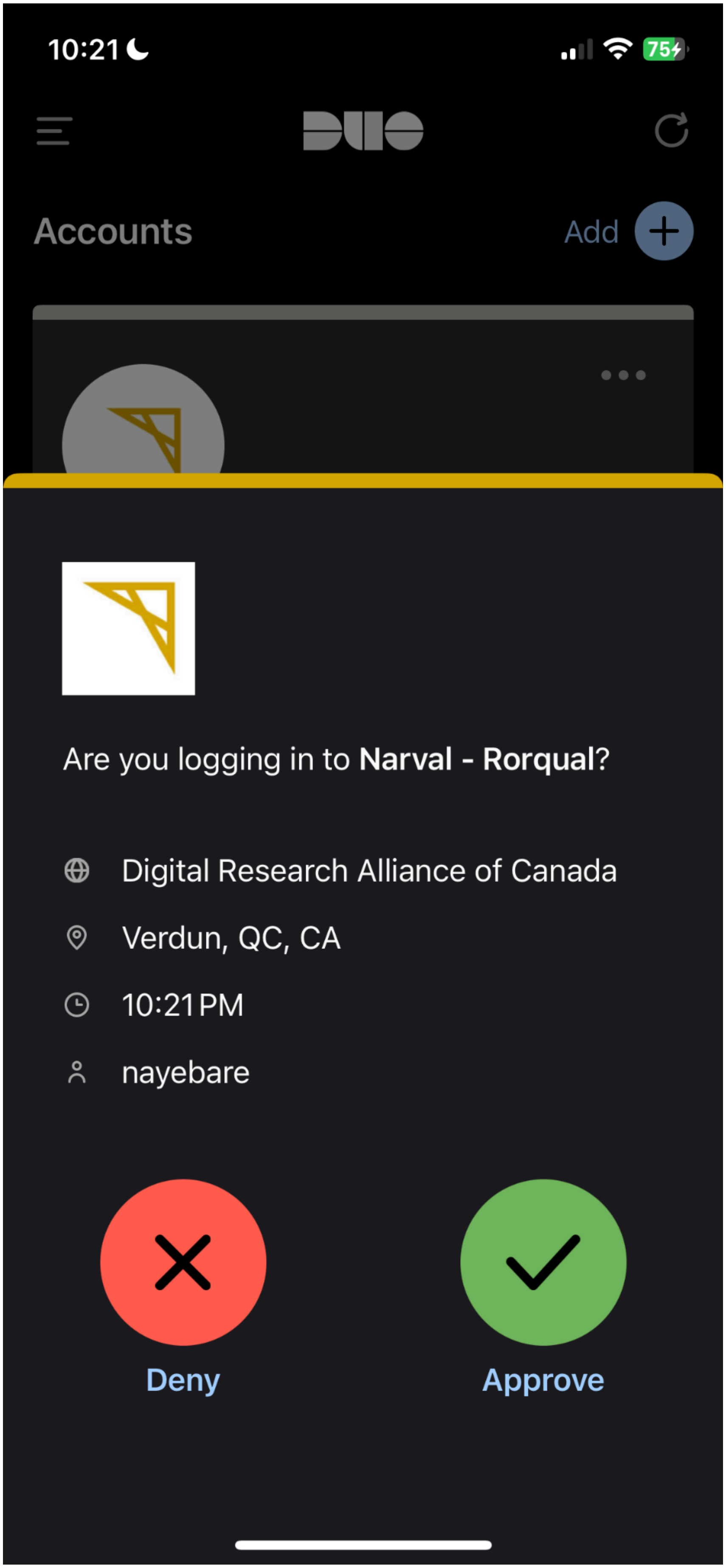




Step 4: Press Details

Step 5: Enter 1 and press Enter.

Step 6: Open the Duo mobile app and tap (approve).


10:21
75
Accounts
Add
Are you logging in to **Narval - Rorqual**?
Digital Research Alliance of Canada
Verdun, QC, CA
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nayebare
Deny
Approve

If this does not work, do it again until it displays your personal environment on Naval.

Step 7: Then, create a new folder and name it, say, Momo_35.

Step 8: Clone the entire code from this GitHub page (https://github.com/tmachtelinckx/halbacharray-GA), **config.py** files, and all into the primary folder.

Step 9: Edit the parameters in the config.py file according to the scanner preferences as shown below.

```
#-----------------------------------RING DIMENSIONS-----------------------------------#

# Define your parameters
InnerBoreDiameter = 80 * 1e-3  # Inner Diameter of the Ring (mm)
OuterBoreDiameter = 160 * 1e-3  # Outer Diameter of the Ring (mm)
magnetSize = 12 * 1e-3          # Length of cube (mm)


#-----------------------------VARIABLE RING PARAMETERS-------------------------------#

amountBand = np.array([1])                # Amount of bands within a ring
bandRadiiGap = np.linspace(0, 0.1, 60)      # Space between Bands (mm)
magnetSpace = np.linspace(0, 0.05, 35)      # Space Between Magnets (mm)
bandSep = np.linspace(0.002, 0.1, 60)       # Space between Bore and 1st band (mm)
```

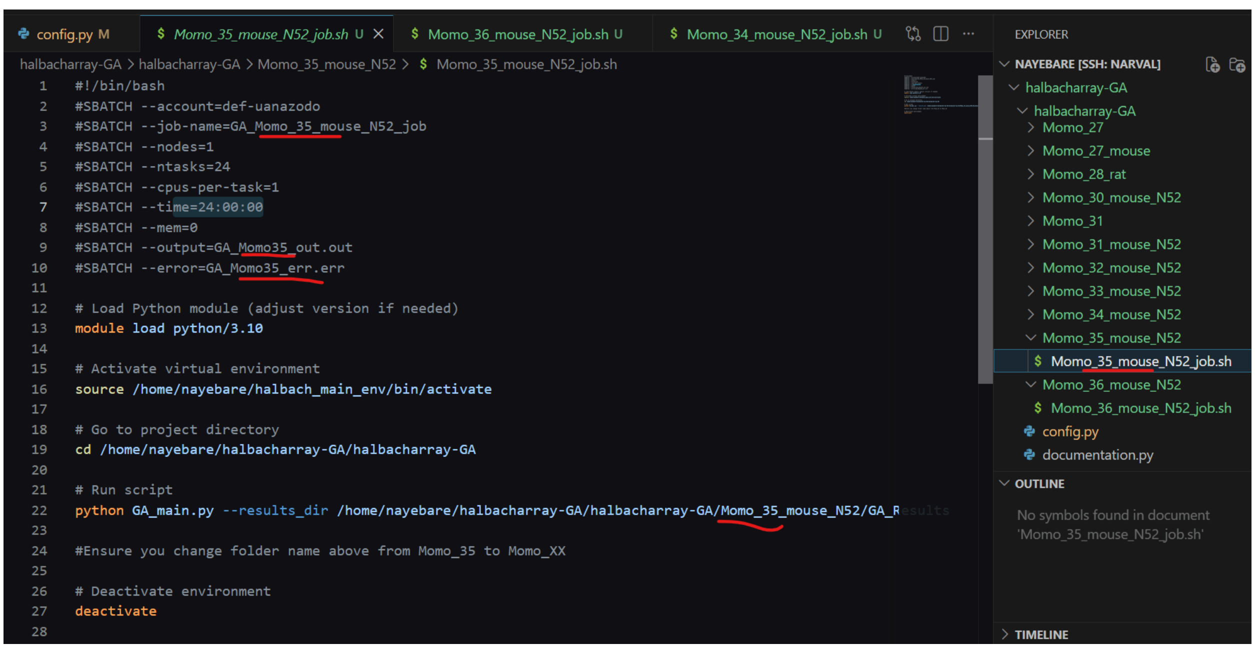
```
ringSep = 0.024
numRings = 7
arrayLength = ringSep * (numRings-1)
ringPositions = np.linspace(-arrayLength / 2, arrayLength / 2, numRings)


#---------------------------- FIELD ERROR PARAMETERS -------------------------------#

# Ensure that homogeneity_weight + field_strength_weight = 1

T_target = 0.05                 # Target field strength in Tesla
homogeneity_weight = 0.9    # Weight for homogeneity error
field_strength_weight = 0.1    # Weight for field strength error
```

Step 10: Edit any other necessary parameters, for example, in the halbachFields.py file, the Bremanence (bRem) value was set to 1.3 for N48 neodymium (NdFeB) magnets.

Step 11: In the Momo_35 folder, create a .job file.

Step 12: In that file, paste all the job file information and edit the time for which the algorithm can run, say 24 hours.

Step 13: Then, in that job file, edit the file name to the initial file name, which is Momo_35 for this case.



Step 14: Press the + button at the bottom to create a terminal.

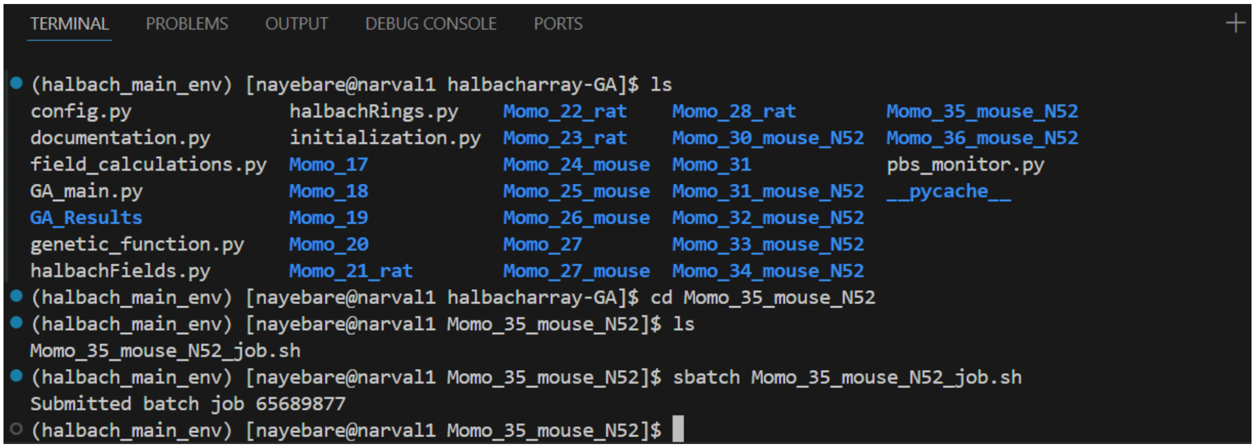
Step 15: ls, enter to see which location you are in.

Step 16: Right-click on the Momo_35 folder and copy the path.

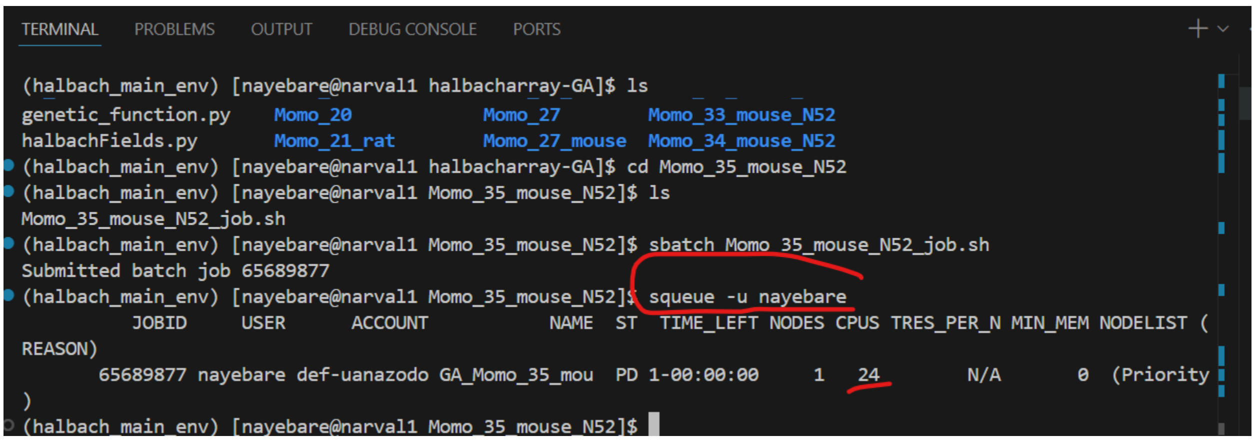
Step 17: In the terminal, write cd space and then paste the folder path

Step 18: Then ***ls*** the components and make sure you see the Momo_35.job.sh file

Step 19: sbatch_ job and press enter.

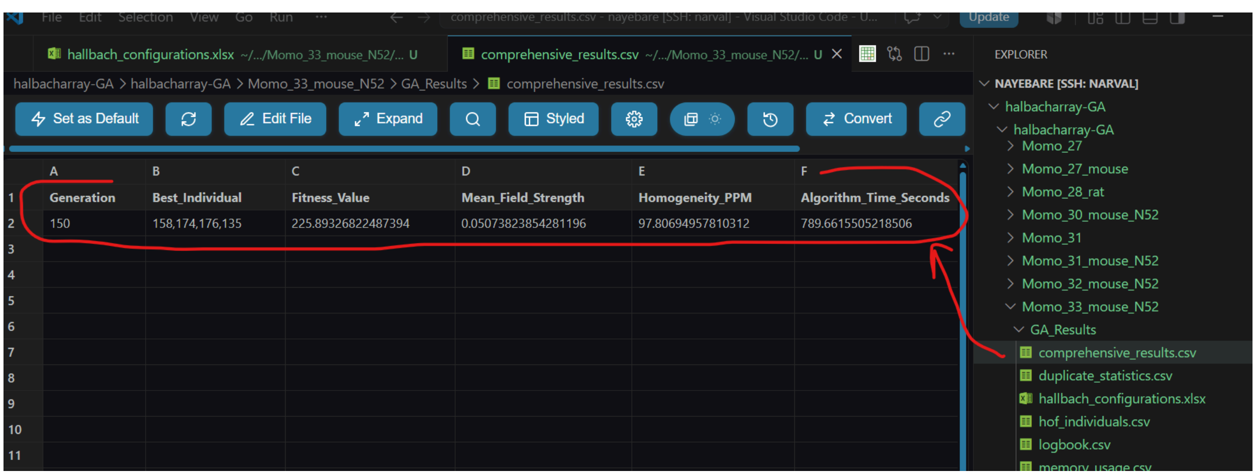


Step 20: To check if the job run, run squeue_nayebare and enter.

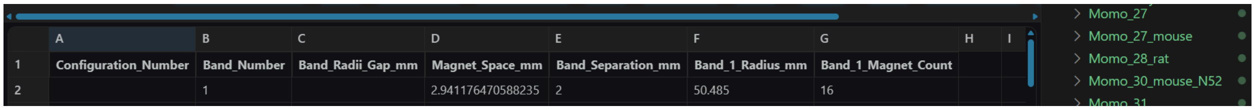


Step 21: If the job finishes, then download the relevant files.

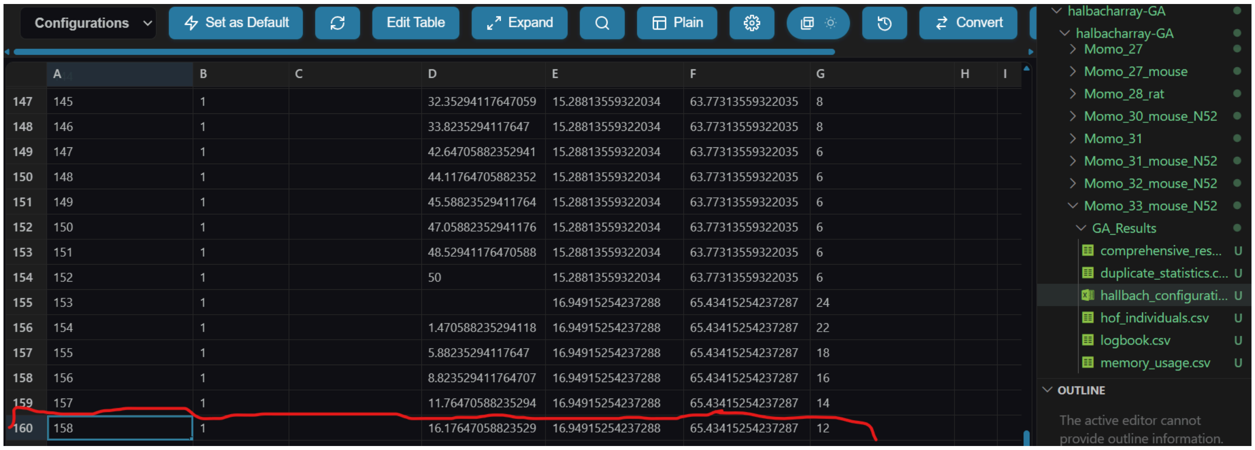
For example, the comprehensive_results.csv file contains the best rings. To look at their specifications, open the halbach_configurations.xlsx file



This is the output in the halbach_configurations file.



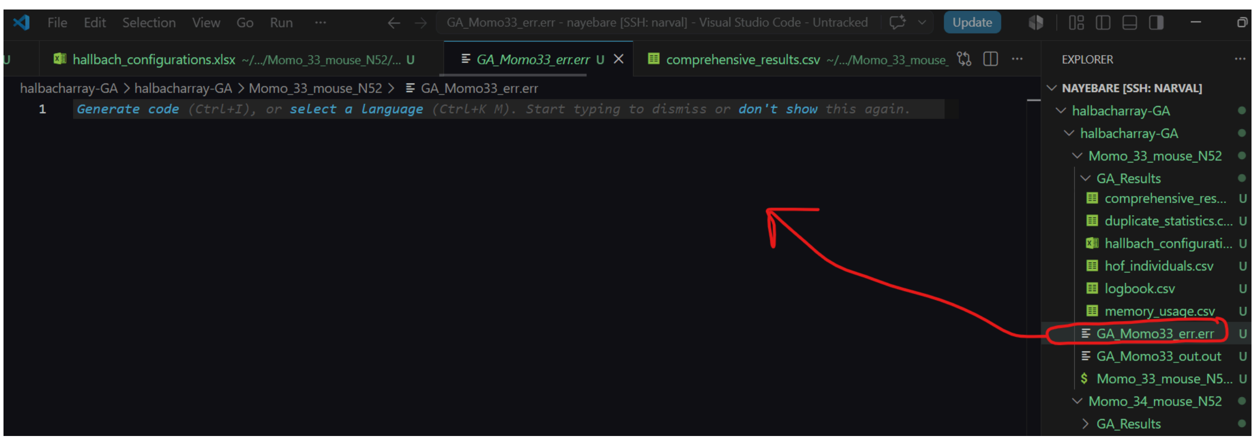
	A	B	C	D	E	F	G	H	I
1	Configuration_Number	Band_Number	Band_Radii_Gap_mm	Magnet_Space_mm	Band_Separation_mm	Band_1_Radius_mm	Band_1_Magnet_Count		
2		1		2.941176470588235	2	50.485	16		



	A	B	C	D	E	F	G	H	I
147	145	1		32.35294117647059	15.28813559322034	63.77313559322035	8		
148	146	1		33.8235294117647	15.28813559322034	63.77313559322035	8		
149	147	1		42.64705882352941	15.28813559322034	63.77313559322035	6		
150	148	1		44.11764705882352	15.28813559322034	63.77313559322035	6		
151	149	1		45.58823529411764	15.28813559322034	63.77313559322035	6		
152	150	1		47.05882352941176	15.28813559322034	63.77313559322035	6		
153	151	1		48.52941176470588	15.28813559322034	63.77313559322035	6		
154	152	1		50	15.28813559322034	63.77313559322035	6		
155	153	1			16.94915254237288	65.43415254237287	24		
156	154	1		1.470588235294118	16.94915254237288	65.43415254237287	22		
157	155	1		5.88235294117647	16.94915254237288	65.43415254237287	18		
158	156	1		8.823529411764707	16.94915254237288	65.43415254237287	16		
159	157	1		11.76470588235294	16.94915254237288	65.43415254237287	14		
160	158	1		16.17647058823529	16.94915254237288	65.43415254237287	12		

Note: In case an error is incurred, check to ensure that the parameters entered in the configuration file are realistic.

The error message would be displayed here:



In that case, an output is not obtained. Edit the scanner parameters in the config.py and submit a job.